

Conditions for coherent vibrations in the cytoskeleton

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Abstract

Mechanism of organization and order in biological systems is not satisfactorily explained yet. Biophysical theories predicted that coherent endogenous electric field of high frequency can play a significant role in organization. The polarity of vibration structures, spectral energy transfer caused by nonlinearities, and energy supply can lead to energy condensation and excitation of coherent states. These conditions are satisfied in the polymer structures of the cytoskeleton, in particular, in the microtubules as follows from experimental findings. Nonetheless, experimental verification of energy condensation and coherent vibrations in the cytoskeleton is still missing. © 1999 Elsevier Science S.A. All rights reserved.

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1. Introduction

Biological activity of living systems manifests a high degree of cooperation and organization. An inconceivable number of various macromolecules, polymers, and organelles interact, join together in chemical reactions, and form larger structures whose properties cannot be predicted from the characteristics of the components. All the processes of biochemical, biophysical, and biological nature are coordinated simultaneously and successively in time but the mechanisms of order and self-organization are largely unknown.

Living matter on the cellular, subcellular and molecular levels does not have spatial order typical for symmetrical structures of inorganic crystalline substances. The ‘translation symmetry’ of microtubules and actin filaments seems to be exceptional. Coordination of biochemical and biophysical processes, dynamic organization and coherence is fundamental to the self-organization of biological systems [1–3]. Self-organization principles are important for evolution and for existence of living matter.

The idea that self-assembly and order are based on biophysical mechanisms using energy is now generally accepted [4,5]. Nevertheless, the concrete mechanism seems

still to be rather obscure. The biological matter has extraordinary polar and polarization properties which suggests that endogenous electric and electromagnetic fields can play a fundamental role in organization and order. Supply of metabolic energy is an important feature which together with nonlinear properties enable explanation of coherent states in biological systems as was done, e.g., by Fröhlich [6–9]. Experimental research on cytoplasmic structures disclosed a fundamental role of cytoskeleton in biological activity of cells. We will compare the properties of the cytoskeleton in living cells with the theoretical requirements for excitation of coherent states.

2. Fundamentals of the Fröhlich theory

In the original work, Fröhlich postulated that long-range quantum mechanical phase correlations exist in biological systems, that the majority of the degrees of freedom behave like being close to thermal equilibrium, that a few modes of motion are very strongly excited and remain far from thermal equilibrium, and that the strong polar character of biological objects suggest longitudinal oscillations as stabilising modes [6]. He deeply analyzed the physical properties of biological systems and developed theory of energy condensation [7–9]. In short, Fröhlich assumed that some modes of ubiquitous thermal vibrations in polar

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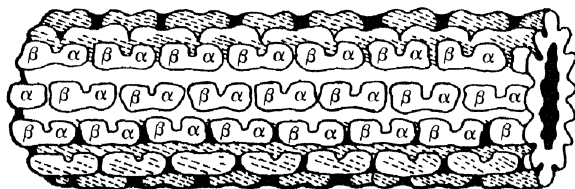


Fig. 1. A schematic figure of a microtubule with tubulin heterodimer molecules in parallel rows (protofilaments). After Refs. [10,12].

systems are excited far from thermodynamic equilibrium. The Fröhlich theory of coherent vibration states has three foundation stones: polar properties of the biological structures, energy transfer between vibration modes (spectral energy transfer), and energy supply to the system.

Long-range interactions within the vibration system as well as between the vibration system and its surroundings (i.e., the heat bath) depend on the electric nature of the vibrations that generate the oscillating electric field. The system may consist of N identical polar oscillators interacting with each other which results in a band of normal modes with frequencies $\nu_1 < \nu_2 \dots < \nu_N$. Fröhlich postulated the rate equation in the form:

$$\begin{aligned} \dot{n}_i = & s_i - \Phi_i [n_i \exp(\beta \nu_i) - (n_i + 1)] \\ & - \sum_j \chi_{ij} [n_i (n_j + 1) \exp(\beta \nu_i) \\ & - (n_i + 1) n_j \exp(\beta \nu_j)], \end{aligned}$$

where $\dot{n}_i$ is the time derivative of the occupation number n_i of the normal mode with frequency ν_i , s_i is the number of energy quanta supplied from metabolic sources in the unit time, and Φ_i and χ_{ij} are the linear and the nonlinear transition probability of energy transfer, respectively. The nonlinear interaction makes possible energy transfer between two normal modes with different frequencies in the vibration system with participation of the heat bath. Energy transfer of this type is a fundamental property of the Fröhlich system and a necessary condition for energy condensation. If energy is not supplied (i.e., $s_i = 0$) the system is in thermodynamic equilibrium. In this case, the third term on the right hand side of the rate equation (the χ term) equals zero as well as the second term (the Φ term) and, therefore, $\dot{n}_i = 0$. The steady state solution of the rate equation (for $s_i \neq 0$), given by the relation:

$$n_i = A \frac{1}{\exp[\beta(\nu_i - \mu)] - 1},$$

where A is the amplitude and $\beta = h/kT$ has the usual meaning, shows that if the excitation potential μ depending on the excitation is high enough, the mode with the lowest frequency is highly excited.

The Fröhlich theory shows the possibility of energy condensation and coherence in the momentum space.

3. Electric properties of the cytoskeleton

The cytoskeleton is an important polymer structure for organization of the cell. The cytoskeleton is composed of intermediate filaments, actin filaments, microtubules, and associated proteins. The microtubules have a fundamental role in organization of the cytoskeleton. The microtubule is a cylindrical hollow tube with the outer diameter of about 25 μm composed of 13 (or 14) protofilaments which are one dimensional chains of heterodimer tubulin molecules (Fig. 1). Description of the cytoskeleton may be found, e.g., in Refs. [4,10–22]. The tubulin molecule is formed by two globular proteins— α - and β -tubulins. The heterodimer molecules can have two conformation states—nontilted and tilted by 29° from the microtubule axis (Fig. 2). The heterodimer molecule is an electric dipole. The microtubules are polar structures with the plus and the minus end embedded in the centrosome—a center of the cytoskeleton. Polarity of the microtubules are closely connected with their function. Fig. 3 shows positions of heterodimers on the wall of a microtubule (a), orientation of the dipole moments in the ferroelectric state—nontilted dimers (b), and the dipole moments of a structure with nontilted and tilted dimers (c). The function of the microtubules depends on phosphorylation and dephosphorylation of microtubule associated tau proteins [16].

4. Nonlinear properties

Vibrations in polar systems excite polarization waves that generate electromagnetic field (with the dominant near zone field electric component) around the vibration structure [5,23]. Energy is exchanged between the near zone field and the vibration structure. (If the vibrations are periodic the energy exchange is periodic too.) A part of the energy may be lost in the ambient medium. Conductivity losses cause ordinary damping represented by the linear transition probability (Φ) in the Fröhlich rate equation. Another type of interaction with the surroundings may exist. A photon of the field can excite a valence electron which loses a part of the energy in a collision (without emission) and then the electron reemit the remaining energy. The collision can increase the energy of the electron too. The emitted quantum of energy differs from the absorbed quantum. This mechanism enables energy trans-

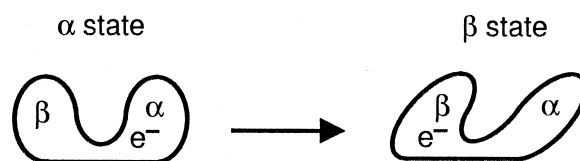


Fig. 2. Conformation change of a tubulin heterodimer from the α state to the β state. The symbol e^- indicates the position of mobile electron charge.

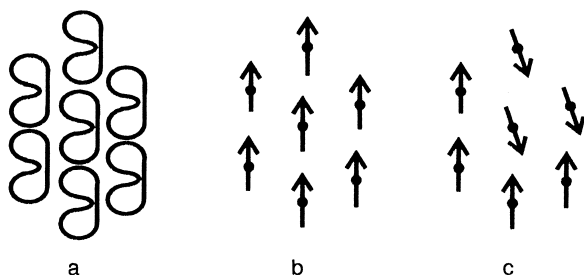


Fig. 3. Position of heterodimers on the wall of a microtubule (a), ferroelectric state of a part of the wall with nontilted heterodimers (b), and dipole moments of a structure with nontilted and tilted heterodimers (c).

fer between modes with different frequencies. Therefore, the polar vibrations in microtubules and actin filaments may be involved in the Fröhlich nonlinear energy transfer.

The microtubule structure is very likely nonlinear by itself. The binding energy of the tubulins in a heterodimer is very likely larger than that between two heterodimers. The elastic force constants along the microtubule chain are not equal and, therefore, the potential as a function of distance of the tubulin ‘particle’ is asymmetric with respect to the equilibrium position. The nonlinear systems can generate higher harmonic components and components with combination frequencies and enable spectral energy transfer too.

5. Vibrations in microtubules and energy supply

The heterodimers in a protofilament form a one dimensional chain. The globular tubulins in the chain may be considered as mass units with translation symmetry (Fig. 1). These mass units have internal vibrations (exerted by atoms and large parts of tubulins) as well as external vibrations in the chain (i.e., vibrations exerted by each tubulin as a whole). A similar rationale may be applied to the actin filaments. We have analyzed the external vibrations in the protofilaments and in the actin filaments using a method similar to that employed in solid state physics [5,24]. Frequency of vibrations have two branches: the optical and the acoustical branch. The highest frequencies of external vibrations are in the range 0.1–1 GHz [5,23,25]. The frequencies of internal vibrations may be much higher (in the submillimeter and far infrared regions).

The dynamics of the microtubules is very likely one of their most prominent features. The microtubules are capable of rapid polymerization and depolymerization which enables exchange of the subunits between soluble and polymer pool. A schematic picture of a microtubule structure with centrosome in the center is shown in Fig. 4. A microtubule grows towards the cell periphery and then shrinks back towards the centrosome. It may shrink partially and grow again or it may disappear completely to be replaced by new microtubules. This process is called the

dynamic instability. The treadmilling (i.e., addition of the tubulin subunits at the plus end and removal of the tubulin subunits at the minus end) appears if both the ends of a microtubule are exposed. After polymerization the guanosine triphosphate (GTP) bound to the β -tubulin is hydrolyzed to the guanosine diphosphate (GDP). Depolymerization and repolymerization is a continual process in animal cells. A similar process of hydrolyzation of adenosine triphosphate (ATP) exists in actin filaments which are important parts of the cytoskeleton too.

Due to depolymerization and repolymerization, the majority of the heterodimer subunits (which are energetically exhausted) is replaced by the energy-rich subunits within about 20 min. The exchange of subunits is much faster in the M phase of the cell cycle. This exchange supplies the standard amount of energy. The dynamic instability provides additional energy supply above the standard value.

We will assess the amount of energy supplied by hydrolysis of the GTP's (attached to the β -tubulins) to the GDP's. The measurements of the hydrolysis of the GTP analogue guanylyl-(a,b)-methylene-diphosphonate (GMPCPP) in solution, in tubulin heterodimers, and in tubulin subunits in the microtubule led to a conclusion that the difference in the free energy was -5.18 kcal/mole, -3.79 kcal/mole, and $+0.7$ kcal/mole [19,26,27]. The reaction with the positive difference of the free energy is thermodynamically unfavorable. Nevertheless, this reaction will ultimately be favorable in the cell. As inorganic phosphate is buffered in the cytosol at approximately 1 mM, then for the equilibrium constant of the reaction smaller than 1, the energy released is -3.38 kcal/mole [27]. A part of the energy is expended on dissociation of the inorganic phosphate but a considerable part is stored in the microtubule lattice (1.7 kcal/mole).

6. Energy usage

Usage of the hydrolysis energy is not well understood. The energy can change bond strengths [13,28] or excite

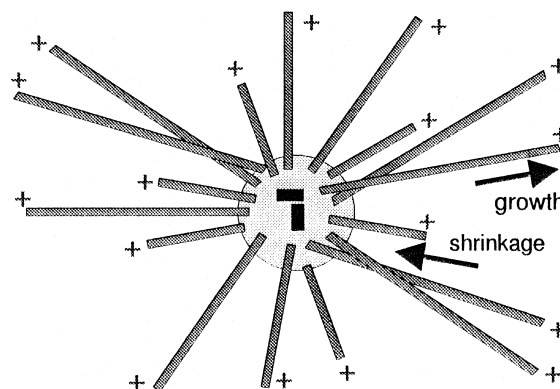


Fig. 4. A centrosome with attached microtubules. The arrows indicate growth and shrinkage in the microtubule array.

Table 1

Theoretical requirements and cytoskeleton properties

Theoretical requirements	Properties of the cytoskeleton
Polar systems	Tubulin heterodimers and actin molecules are dipoles
Long-range interaction	Interaction by the electric field
Spectral energy transfer by	Mediated by
(a) nonlinear interaction with a heat bath	(a) a near zone electric field
(b) nonlinear intrinsic properties	(b) nonlinear oscillations in asymmetric potential valleys
Energy supply	Hydrolysis of
	(a) GTP in microtubules
	(b) ATP in actin filaments
<i>Consequences</i>	
Energy condensation	Not experimentally proved
Coherence	

soliton or soliton-like waves [11,28]. It can be used for conformation changes of tubulin heterodimers (or for imparting energy via tubulin conformation changes) [12] and for kink wave reorientation of heterodimer dipoles [11,13]. It can excite coherent lattice vibrations by the Fröhlich mechanism [13,28] and/or by the intrinsic nonlinear mechanism. Real physical mechanisms of energy transformation are irreversible and a part of the energy is always transferred to vibrations. Even if the energy levels of two conformation states of a tubulin subunit are different (e.g., if the level of the β state is higher than that of the α state) after transition to the higher energy level, a part of the energy which is used to surmount the potential barrier between the conformation states can be transformed into vibration energy. (The opposite relaxation process should convert energy to vibrations too.) Creation of the tense state with altered forces between the subunits in the microtubule has to supply energy into the lattice. Therefore, we may assume that energy is stored in the microtubule lattice not only in the conformation changes and in the tense states of the lattice but also in the vibration states [5,25].

The energy stored in the microtubule can do work in the cell but a precise manner in which the energy is used is still not understood. We assume that the energy can excite the lattice vibrations and polarization waves, which may be coherent on account of the Fröhlich mechanism or of the intrinsic nonlinear mechanism. The majority of the energy stored in the microtubules may be converted into the vibrations. Therefore, the vibrations are excited and can be far from thermodynamic equilibrium. The electric field generated by the polar vibrations can do work in the surroundings of the microtubule. In the vicinity of the microtubule (e.g., at a distance of about 10 nm) the intensity of the electric field is about 10^5 V/cm (the vacuum value) [5,23].

We assessed the amount of energy which might be supplied to the cytoskeleton vibrations from the hydrolysis of GTP. If the energy supplied by the hydrolysis of GTP is 1.7 kcal/mole, then the energy per one entity is about three times greater than its energy in thermodynamic equilibrium

in the frequency range from 10^7 to 10^{11} Hz. The standard amount of energy is supplied within about 20 min when the microtubule pool of heterodimers is exchanged. Due to dynamic instability, the total amount of energy supplied to the microtubule structure in this time interval is considerably greater.

7. Theoretical requirements and microtubule properties

Table 1 shows comparison of theoretical requirements for creation of coherent states and properties of the microtubules. The microtubules satisfy the conditions for energy condensation and coherence. The measured electromagnetic activity of living cells (observed, for instance, in yeast cells—*Saccharomyces cerevisiae* [29,30]) may be assigned to microtubules. Further measurements have to be carried out to prove or disprove coherence of the emitted electromagnetic field.

8. Conclusions

Microtubules are polar—their heterodimer tubulin subunits are electric dipoles whose near-zone field mediates interaction with the ambient medium. The microtubule structure is nonlinear. Energy supply is provided by the hydrolysis of GTP to GDP. The microtubules in the cytoskeleton satisfy the requirements for excitation of coherent vibrations but at present experimental proof of coherent states is missing. We assume that electromagnetic activity of living cells originates in the cytoskeleton and in particular in the microtubules.

Acknowledgements

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